

cause the Coromandel Express entered a loop line—a section of track built to ease rail operations—and hit a stationary goods train. The derailed carriages of the Coromandel Express then hit the tail-end bogies of the Bengaluru-Howrah Superfast Express that was passing on the down line. “The resulting triple accident is one of the rarest of the rare instances in rail transport,” avers safety expert Lalit Chandra Trivedi, who retired as General Manager of East Central Railways.

Others, however, view this as symptomatic of a larger issue. “Transportation-related accidents in India have a pattern with roots in systemic failure and the railways is also suffering with this problem,” an international rail safety expert tells *BT* requesting anonymity. The person mentions an incident a few years ago when a train derailment near New Delhi was caused after old rails were replaced with tracks that had been used at another railway yard! “Ultimately, it is about both a lack of robust safety design-install-test process, enforcement, staff competency management and accountability.”

Filling up vacancies for engineering roles and putting technology adoption on the fast track are some ways to prevent a recurrence, say experts. “There has been no fresh induction of engineers through engineering services exams in the past few years,” notes another former railway official. “There needs to be enhanced focus on field inspections even while limiting time spent on meetings.”

Seasoned experts like Trivedi attribute the fatal crash to a combination of factors such as underinvestment in safety and the transition of the state-owned behemoth into an enterprise that is only now learning to work closely with the private sector. On the shortage of skilled staff,

SAFETY IN STEPS

Railway experts urge a six-step process to prevent the recurrence of Balasore-like accidents in the future



LEVEL 1

Design-stage check by the contractor



LEVEL 2

Provision for independent checking by a competent person



LEVEL 3

Integration check of systems like signalling, telecom, track, power, etc., by all personnel/contractors



LEVEL 4

Client (Indian Railways) check by a senior engineer



LEVEL 5

Installation check must be done by a different firm or set of people to ensure work has been done as per the approved design



LEVEL 6

Final testing to be done by an independent agency

The above must be carried out by staff who are checked for their competency at least once yearly and issued requisite licence

Source: Industry, *BT* Research

Trivedi says, “While the organisation is otherwise overstaffed, there is a shortage of personnel at a local level in critical areas of safety requiring specialised skills.” Similarly, signals and tracks don’t support speeds higher than 110 kmph, resulting in underutilisation of rolling stock capable of 160 kmph. Consequently, measures at providing 2x25 KV overhead equipment (OHE) cantilever, tracks capable of supporting semi-high-speed trains and implementation of the automatic train protection (ATP) systems such as the indigenously-developed Kavach end up looking half-hearted.

Experts also cite the slow adoption of cutting-edge technologies and a lack of structured R&D activities. “The multiple attempts at introducing technology in the railways haven’t progressed to satisfactory levels yet. That needs to be addressed on a war footing now as there is a lot to be done, especially with regard to Kavach and other proven technologies,” says a senior executive at a leading railway original equipment manufacturer.

In 2019, the government abruptly shelved the roll-out of the European Train Control System, where Bombardier had emerged as the lowest bidder for a deal that was potentially worth around ₹1,500 crore. “Had it not been cancelled, its implementation would have helped the railways and passengers,” says the executive. Sources attribute this to a lack of finances and the hassle of adapting foreign technology to local conditions. “Discussions to give thrust to technology have been going on for over 10 years. It is a no-brainer, deploying technology in a brownfield environment is never easy but then ways have to be found,” says the executive.

“With all this equipment, maintenance and SOPs are very important. We have to ensure that